

Rockborne

Multiple Linear Regression

Fitting a plane through many features

Objectives



Multiple vs Single LR

From a line to a plane.



Establishing Variables

A quick recap.



Equations & Accuracy

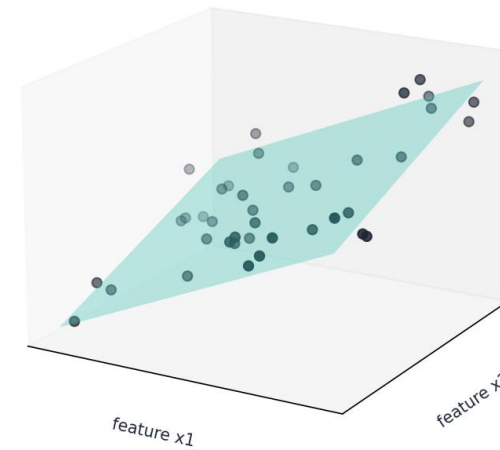
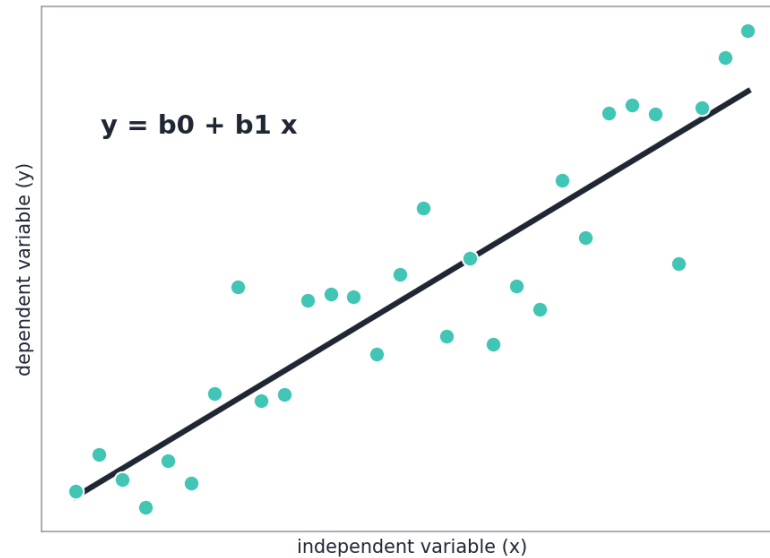
The equation and error metrics.

Introduction

Multiple vs Single Linear Regression

...and why that matters

Multiple vs Single Linear Regression



SLR fits a line to a single feature.

MLR fits a plane (a hyperplane) to many features.

It works with higher-dimensional data.

Same idea, more predictors.

Multiple Linear Regression



Establishing Variables (Recap)

Dependent vs Independent Variables

Dependent variable (y)

- The target we want to explain or forecast.
- Its value changes based on the independent variables.
- Examples: test scores, blood sugar.

Independent variables (x1..xn)

- Impact the dependent variable's output.
- Are not reactive to the dependent variable.
- Examples: time, age, income.

Multiple Linear Regression



Equations and Accuracy

The MLR Equation: $y = b_0 + b_1 x_1 + \dots + b_n x_n + e$



b_0 - Intercept

The baseline value of y .



$b_1..b_n$ - Slopes

One per feature: its effect, holding the others constant.



e - Error term

The residual: what the model does not capture.

Interpreting the Coefficients



Hold others constant

Each coefficient is that feature's effect with the others fixed.



Standardise to compare

Raw coefficients are not comparable across different scales.



Watch multicollinearity

Correlated predictors make coefficients unstable.

Accuracy: RMSE



What it is

The average error the model makes when predicting.



How

The square root of the mean squared error (MSE).



Note

Same units as the target; larger errors are penalised more.

Accuracy: RSE (model sigma)



What it is

RMSE adjusted for the number of predictors.



Rule

The lower the RSE, the better the fit.



Note

Very close to RMSE, especially on large data.

Accuracy: MAE



What it is

The average absolute difference between actual and predicted.



vs RMSE

Less sensitive to outliers (no squaring).



Use it

When large errors should not dominate the metric.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Multiple Linear Regression - Guided Walkthrough



Then: Independent Lab

In this lesson, we explored



Multiple vs Single LR

MLR fits a plane to many features.



Establishing Variables

Dependent (y) vs many independents.



Equations & Accuracy

The equation, coefficients, RMSE /
RSE / MAE.

Digging Deeper?



[LinearRegression \(API\)](#)

scikit-learn



[Linear Models \(User Guide\)](#)

scikit-learn



[Multiple Linear Regression](#)

Statology

Questions?